

研究進捗報告 (2026/02/02)

澁谷優人

データセット

Baltrum_S_FP1
GT付きのピクセル数が2457178
trainに用いたのがその1%
testに用いたのが残り99%
random_seed=42

学習結果比較 (デフォルト vs 提案手法)

全体指標 (Overall Metrics)

Metric	Default	Proposed
OA (Overall Accuracy)	90.09%	88.29%
AA (Average Accuracy)	86.40%	84.40%
Kappa	0.89	0.85

クラス別精度 (Class-wise Metrics)

注: 数値は (Precision / Recall / F1-score) の順で記載

Class ID	Class Name	Default	Proposed
0	Tidal flat	96.78 / 92.67 / 94.68	96.95 / 81.03 / 88.28
1	Water	96.92 / 98.62 / 97.76	93.41 / 98.13 / 95.71
2	Coastal shrub	69.63 / 69.08 / 69.35	69.78 / 50.00 / 58.26
3	Dense, high vegetation	62.58 / 65.36 / 63.94	46.23 / 84.34 / 59.73
4	White dune	74.29 / 76.85 / 75.55	90.93 / 52.62 / 66.66

Class ID	Class Name	Default	Proposed
5	Peat bog	93.25 / 87.49 / 90.28	86.29 / 87.34 / 86.81
6	Grey dunes	84.09 / 87.76 / 85.89	80.05 / 91.34 / 85.32
7	Couch grass	82.09 / 83.94 / 83.00	76.58 / 84.30 / 80.26
8	Upper salt marsh	93.78 / 89.80 / 91.75	91.77 / 89.62 / 90.68
9	Lower salt marsh	93.55 / 96.45 / 94.98	94.07 / 94.87 / 94.47
10	Sand	96.76 / 96.64 / 96.70	95.27 / 94.37 / 94.82
11	Settlement	91.56 / 86.15 / 88.77	90.40 / 81.65 / 85.81

平均 (Averages)

Type	Default	Proposed
Macro Avg	86.27 / 85.90 / 86.05	84.31 / 82.47 / 82.23
Weighted Avg	90.00 / 89.89 / 89.91	88.22 / 87.18 / 87.10

学習結果 写真

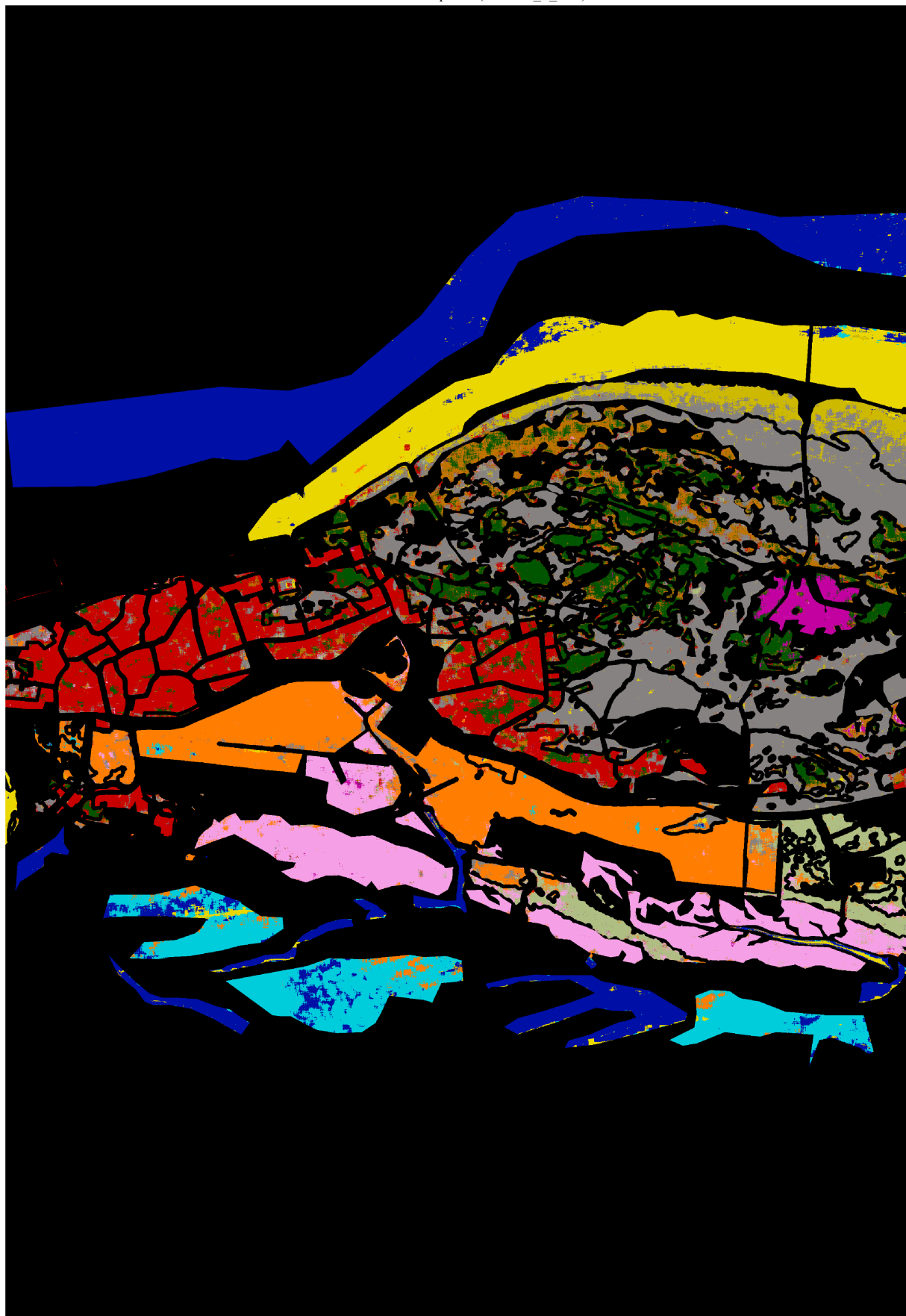
GT

Ground Truth (Baltrum_S_FP1)





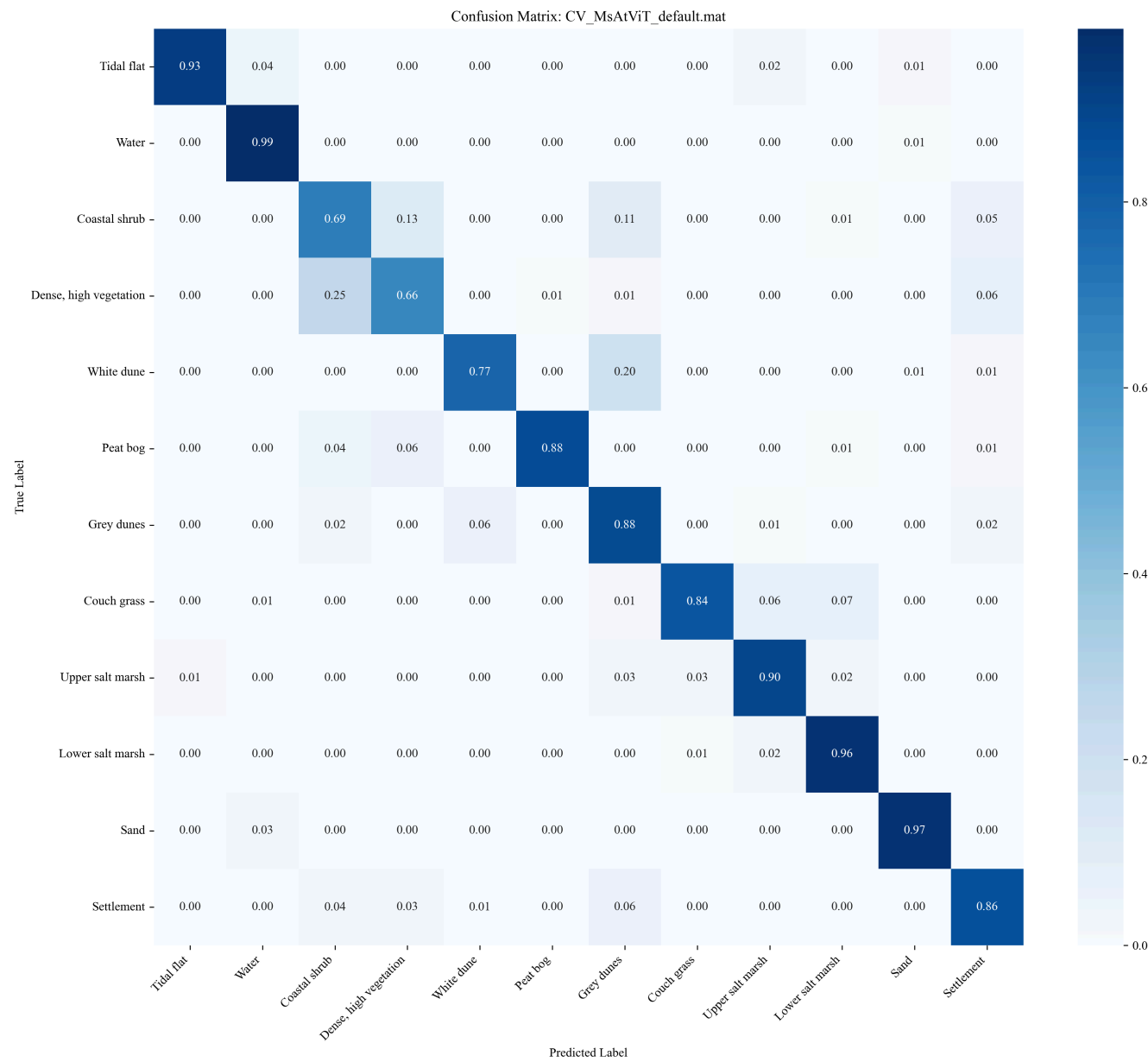
- Tidal flat
- Water
- Coastal shrub
- Dense, high vegetation
- White dune
- Peat bog
- Grey dunes
- Couch grass
- Upper salt marsh
- Lower salt marsh
- Sand
- Settlement



- Tidal flat
- Water
- Coastal shrub
- Dense, high vegetation
- White dune
- Peat bog
- Grey dunes
- Couch grass
- Upper salt marsh
- Lower salt marsh
- Sand
- Settlement

混同行列

先行研究



提案手法

